Software Requirements Specification (SRS)

Invoice Management System(InvoNotify)

Version 1.0 approved

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SOFTWARE REQUIREMENTS SPECIFICATION

for

INVOICE MANAGEMENT SYSTEM

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1. INTRODUCTION

1.1 Objective

The objective of this project is to design an Invoice Management System (IMS) that helps businesses streamline their billing processes. The primary goal is to provide a centralized, automated platform to create, send, and track client invoices, ensuring efficient financial management from anywhere.

1.2 Document Conventions

This document uses standard business documentation formatting. Technical terms like Next.js, Prisma, and Vercel Cron refer to the specific technologies implemented in the system. High-priority items are capitalized for emphasis.

1.3 Intended Audience and Reading Suggestions

The intended audience includes business owners, accountants, and administrative staff who generate bills and track revenue. Developers and project managers should use this document to understand the project's technical and functional scope.

1.4 Project Scope

The Invoice Management System provides a comprehensive tool for digital billing cycles. It eliminates manual paperwork by allowing users to oversee revenue, manage client accounts, generate secure invoices, and track payment statuses (Paid, Unpaid, Overdue) through an intuitive web interface.

1.5 References

Docs which are used to accomplish the given requirements.

- Next.js Documentation (nextjs.org)

- Prisma ORM Documentation (prisma.io)

- Vercel Deployment and Cron Guides (vercel.com)

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2. OVERALL DESCRIPTION

2.1 Product Perspective

As a web-based application, the Invoice Management System provides:

- Invoice Generation: Customizable invoices with tax calculations (GST/CGST/SGST/IGST).
- Payment Tracking: Monitoring of billing lifecycles and revenue collection.
- Automated Reminders: Systematic notification system for pending and overdue payments.

2.2 Product Features

- Secure User Authentication via NextAuth.js.
- Financial Dashboard with real-time analytics.

- Multi-template Invoice Creation (Standard, Professional).

- Automated Email and SMS reminders(trial).

- Recurring overdue alerts based on user-defined intervals.

2.3 User Classes and Characteristics

- Administrators/Owners: Full access to manage billing, configuration, and reports.

- Accountants: Access to draft invoices, verify payments, and manage client data.

2.4 Operating Environment

- Client: Windows, macOS, Linux (Accessible via modern browsers like Chrome).

- Server/Hosting: Vercel Platform.

- Framework: Next.js (TypeScript-based).
- Database: Prisma ORM with PostgreSQL.

2.5 Design and Implementation Constraints

- Deployment: Optimized specifically for the Vercel Platform.
- Scheduling: Automated tasks run via Vercel Cron daily at 09:00 UTC.
- Security: Sensitive API endpoints are protected via CRON_SECRET headers.
- Data Integrity: Managed strictly through Prisma database migrations.

2.6 User Documentation

The project includes a README.md file for installation and environment setup. The web

interface features intuitive guides for configuring automated reminders.

2.7 Assumptions and Dependencies

- Dependent on valid SMTP/Gmail credentials for email dispatch.

- Dependent on Twilio or similar gateways for SMS notifications.

- Reliant on Vercel infrastructure for hosting and task scheduling.

3. SYSTEM FEATURES

3.1 Automatic Invoice Reminders

3.1.1 Description: Sends automated alerts via Email or SMS before/on the due date and at recurring intervals if overdue and if the customer is genuine and Is the regular customer from the last 2-3 years then that customer will be notified automatically and can be manually notified if the client wants to.

3.1.2 Functional Requirements:

- Users can toggle reminders on/off per invoice.
- Choice of delivery channel: Email, SMS, or Both.
- Customizable offsets (7, 3, or 1 day before due date).
- Recurring overdue alerts (e.g., every 3 days).

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4. EXTERNAL INTERFACE REQUIREMENTS

4.1 User Interfaces

- Responsive front-end built with Next.js and Tailwind CSS.
- Optimized typography using the Geist font family for readability.
- Visual data representation using Recharts.

4.2 Hardware Interfaces

- Accessible from any computer or mobile device with a modern web browser and internet connection.

4.3 Software Interfaces

The following table summarizes the core technologies used:

Software Used	Description
OS & Hosting	Vercel Platform (Production) / Multi-OS (Dev)
Database	Prisma ORM with PostgreSQL
Framework	Next.js (TypeScript 98.9% efficiency)
Notification Interface	SMTP/Gmail for Email; Twilio for SMS

Task Scheduler | Vercel Cron Jobs(Daily at 09:00 UTC)

4.4 Communications Interfaces

- HTTPS for secure web communication.

- SMTP/GMAIL-Playground for outbound email delivery.

- RESTful APIs for internal and external service integration.

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5. OTHER NONFUNCTIONAL REQUIREMENTS

5.1 Performance Requirements

- High responsiveness and low latency for all financial operations.
- Efficient processing of bulk reminders during the morning cron window.

5.2 Safety Requirements

- Prevention of accidental spam through strict reminder logic.
- Protection of sensitive billing data from unauthorized external access.

5.3 Security Requirements

- Strict endpoint protection using CRON_SECRET environment variables.

- Secure session management preventing unauthorized dashboard access.

5.4 Software Quality Attributes

- Availability: Guaranteed 24/7 access via Vercel's global infrastructure.

- Correctness: High precision in GST calculations and schedule adherence.

- Maintainability: Robust TypeScript codebase with automated migrations.

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6. OTHER REQUIREMENTS

- Database migrations must be executed via 'pnpm prisma migrate deploy'.
- Proper environment variable configuration is mandatory for all production features.

APPENDIX A: GLOSSARY

- Prisma: Object-Relational Mapper (ORM) for secure database access.
- Vercel Cron: Scheduler for automated background tasks.
- Next.js: Core framework for UI and API routes.

- TypeScript: Language used for type-safe code implementation.

APPENDIX B: ANALYSIS MODELS

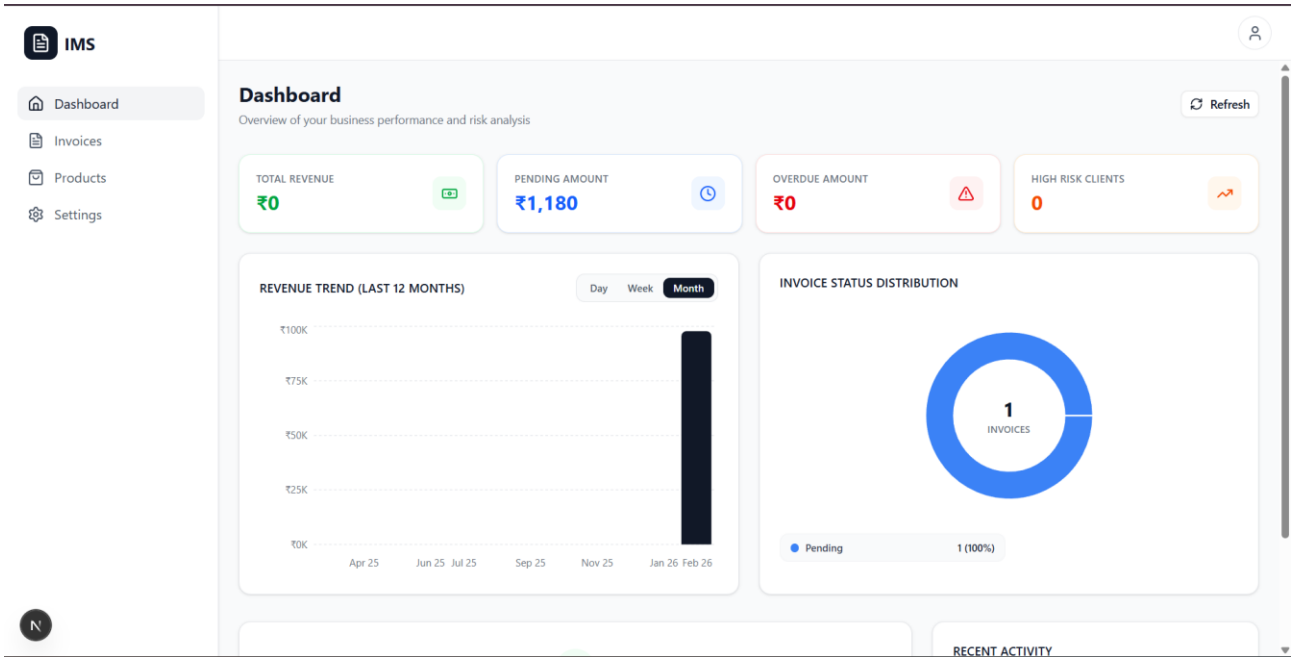
- Data Flow: User creates invoice -> Data persists in PostgreSQL -> Cron triggers dispatch -> Notification sent via SMTP.

APPENDIX C: ISSUES LIST

- Integration of regional SMS providers.

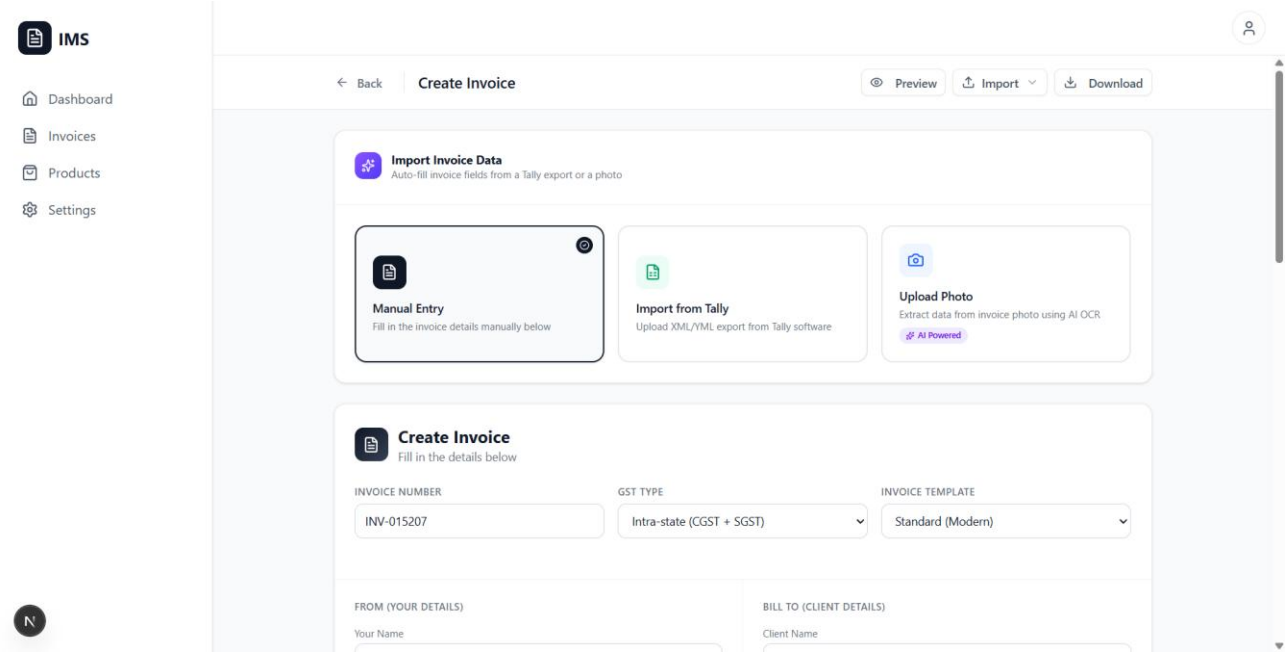
- Future support for direct one-click payment links (Stripe/Razorpay).

- Dashboard with revenue analytics and status overviews.

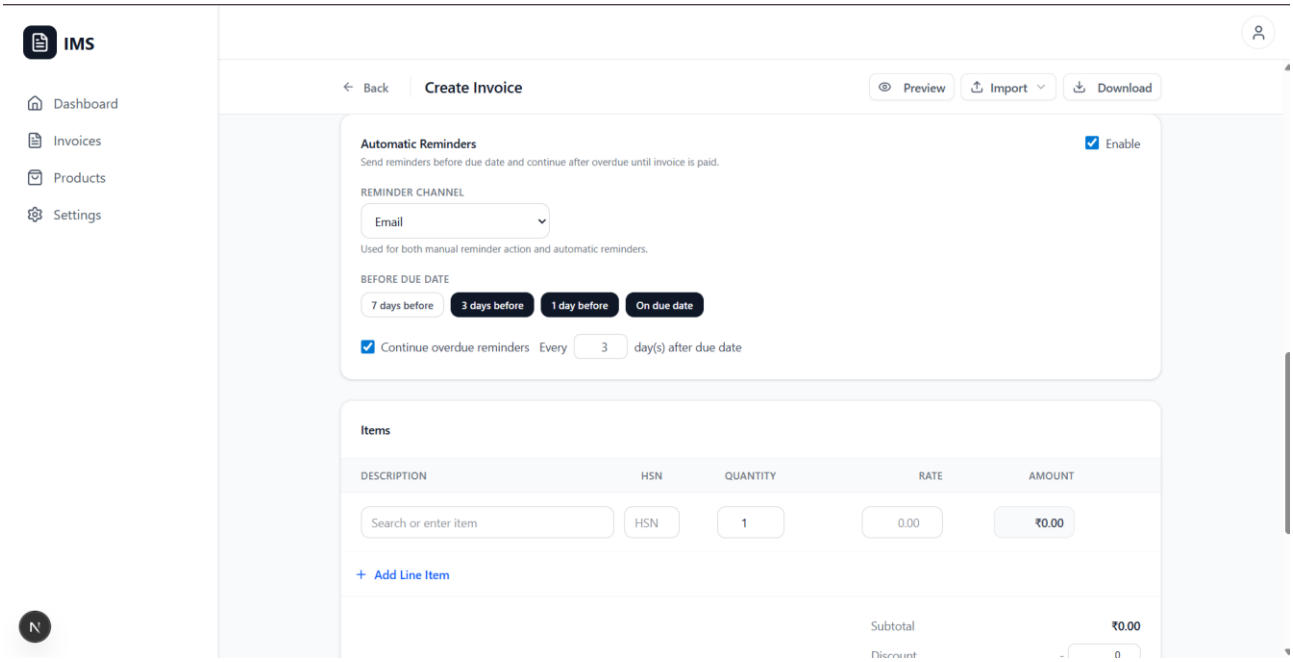


- Invoice creation with multi-template support (Standard, Professional, etc.).

(to be added by the next milestone)

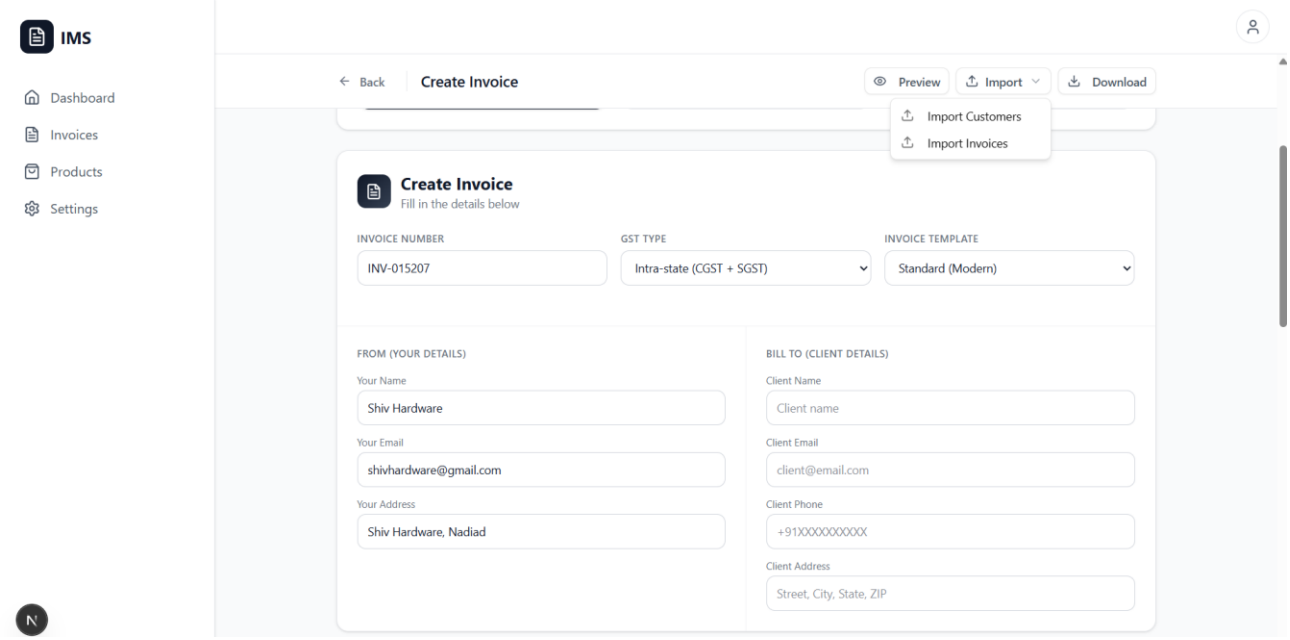


- Automated reminder configuration with custom offsets (e.g., 7, 3, 1 day before due).



- Recurring overdue alerts.

- Client and Product management for rapid invoice drafting.



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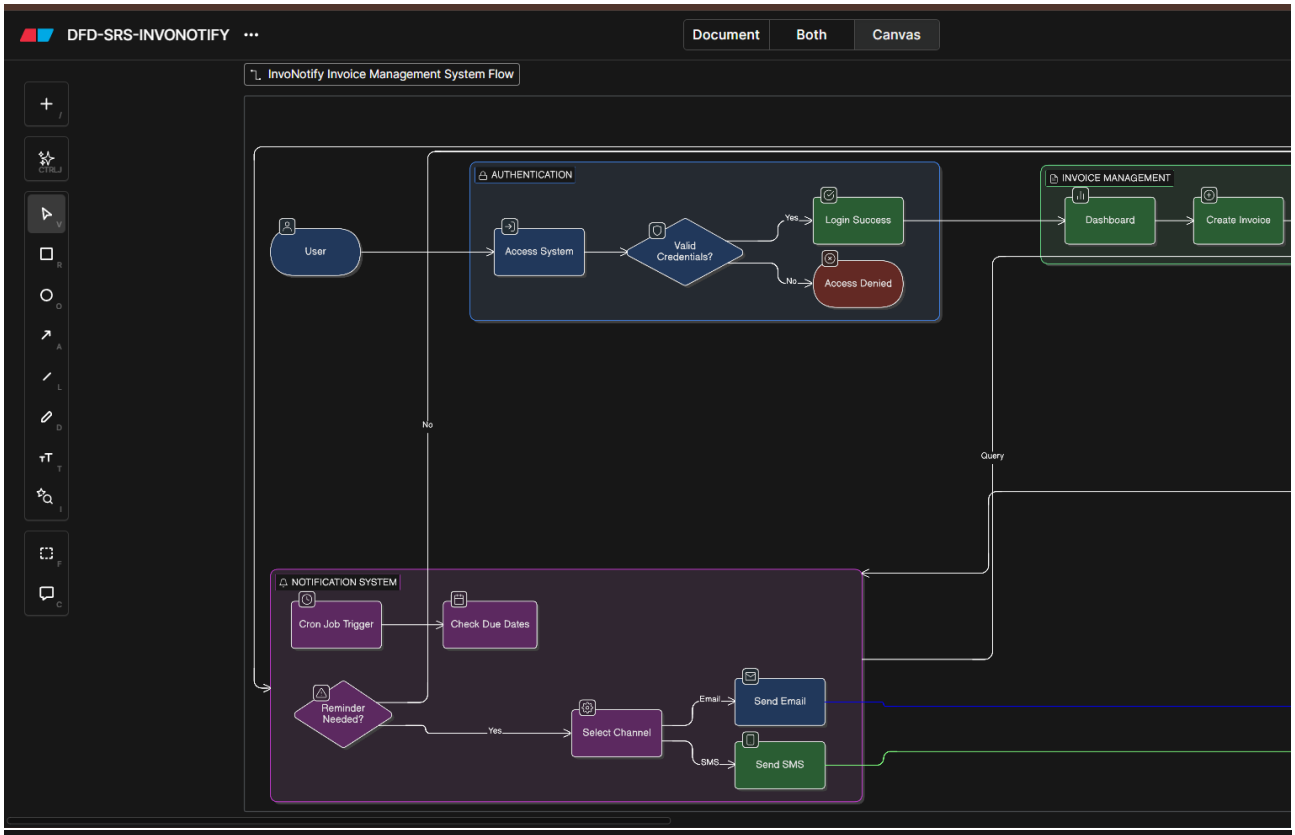
DFD-SRS-INVONOTIFY
Document
Both
Canvas
InvoNotify Invoice Management System Flow
AUTHENTICATION
User
Access System
Valid Credentials?
Yes
No
Login Success
Access Denied
INVOICE MANAGEMENT
Dashboard
Create Invoice
No
Query
NOTIFICATION SYSTEM
Cron Job Trigger
Check Due Dates
Reminder Needed?
Yes
Select Channel
Email
SMS
Send Email
Send SMS

DFD-SRS-INVONOTIFY
Document
Both
Canvas
Ctrl K
Share
62%
Process Complete
Create Invoice
Select Template
Add Client Details
Add Line Items
Calculate Tax
Save Invoice
DATA LAYER
PostgreSQL Database
Invoice Records
Client Records
PAYMENT TRACKING
Check Payment Status
Invoice Paid?
Yes
No
Mark as Paid
Mark as Unpaid or Overdue
Notification
Notification
Client
Notification Sent